Mohith Thuraiyur Kirubanandham

MSc Data Science student, University of Surrey

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Profile

MSc Data Science student at the University of Surrey (2026–2027) with a BSc in Artificial Intelligence and Data Science. I build machine learning, deep learning and IoT projects in Python, from misinformation detection and stroke-risk prediction to REST microservices, an encrypted vehicle black box and a market drawdown alert system. I am looking for opportunities in data science, machine learning and AI research where I can apply statistical modelling to real-world problems.

Education

MSc Data Science, University of Surrey, Guildford, UK · 2026 – 2027 (in progress)

- Focus: advanced data science, machine learning, AI, statistical modelling, predictive modelling, data-driven research

BSc Artificial Intelligence and Data Science, Rathinam College of Arts and Science, Coimbatore, India · 2023 – 2026

- Coursework: Python, deep learning, data analysis, DBMS, data visualisation, IoT, predictive analytics

IELTS: 6.5

Projects

Asynchronous Analytics Job API, Python, Flask, REST · GitHub

- Built a REST API that accepts analytics jobs, processes them in background threads and returns 202 Accepted with a job ID that clients poll for status (QUEUED, RUNNING, SUCCEEDED).
- Added request validation with structured JSON errors (400 for bad input, 404 for unknown jobs, 409 when results are not ready), plus health and service-metadata endpoints.
- Designed the job interface around MapReduce and Spark engine options; processing is simulated in this coursework version.

Bookshop Microservices API, Python, Flask, SQLite, REST · GitHub

- Built a three-service system of an API gateway, a catalogue service and an orders service, each configured through environment variables, with SQLite storage in the catalogue and orders services.
- Gateway routes public API calls to internal services, aggregates service health checks, applies request timeouts and returns 502 when a downstream service is unavailable.
- Orders service checks and reduces stock through HTTP calls to the catalogue service, returning 409 on insufficient stock and 503 when the catalogue is down.

RK Creators and Builders Website, HTML, CSS, JavaScript, Netlify Functions, Node.js · GitHub

- Built a multi-page website and admin panel for a family-run construction business in Tamil Nadu, covering services, process, projects, about and contact pages.
- Wrote serverless functions for enquiry submission (honeypot spam check, per-IP rate limiting, validation, email via Nodemailer), admin login, and project and enquiry management stored in Netlify Blobs.
- Set up a GitHub Actions workflow that deploys the site to Netlify on every push to main.

NIFTY 50 Drawdown Alert System, Python, yfinance, SQLite, Docker, Telegram Bot API · GitHub

- Built a monitoring service that tracks the NIFTY 50 against its rolling 52-week high and sends Telegram and email alerts at configurable drawdown levels (10% to 50%).
- Stored alert state in SQLite so each threshold fires once and restarts never resend alerts; added stale-price rejection and market-hours and holiday gating.
- Packaged for Docker Compose with pytest tests and a companion Telegram assistant bot for status queries. Monitoring only, with no trading access.

Secure Vehicle Black Box and Forensic Monitoring, NodeMCU (ESP8266), Arduino C/C++, Python, ThingSpeak · GitHub

- Built an IoT prototype that reads accelerometer/gyroscope (MPU6050), GPS (NEO-6M) and temperature/humidity (DHT11) sensors and encrypts each data packet with RC6.
- Wrote Python scripts to poll the device, log encrypted records to CSV and decrypt them for post-incident analysis, with ThingSpeak cloud dashboards.
- Designed to flag unrealistic sensor values and log tampering to support accident investigation and insurance checks.

Fake News Detection Model, Python, CNN, NLP

- Developed a convolutional neural network that classifies news text as real or fake, including data preprocessing and text classification.

Stroke Prediction System, Python, scikit-learn

- Built a machine learning model that estimates stroke risk from patient attributes, covering exploratory data analysis, feature selection and predictive modelling. Academic project, not for clinical use.

Medical Inventory Management, Salesforce, Apex · GitHub

- Team project (Naan Mudhalvan) building a Salesforce Lightning app to manage suppliers, products, purchase orders and expiry dates, using custom objects, validation rules, profiles, flows and an Apex trigger that totals order items per purchase order.

Technical skills

- **Programming:** Python, R, C++, JavaScript, SQL (MySQL), Apex
- **Backend and web:** Flask, REST API design, microservices, Node.js, Netlify Functions, HTML and CSS
- **Machine learning:** scikit-learn, CNNs and deep learning, NLP text classification, exploratory data analysis, feature selection, predictive modelling
- **Data and tools:** Power BI, pandas, SQLite, Docker, Git and GitHub, GitHub Actions, pytest
- **IoT:** NodeMCU (ESP8266), Arduino, sensor integration, ThingSpeak, RC6 encryption
- **Professional:** teamwork, communication, commitment, organisation

Languages and interests

- English (fluent), Tamil (fluent)
- Running, trekking and bodybuilding, which have built the discipline and resilience I bring to my studies